

6.2 BASIC EMPTY MASS CENTER OF GRAVITY

The procedure for establishing mass and moment (relative to the reference datum) of the empty helicopter is described in the BO 105 Maintenance Manual (MM). The MASS AND BALANCE RECORD (Form MBR-1) appended to this Section is used to maintain a continuous history of changes to the basic “as delivered” helicopter mass and balance data.

6.2.1 Mass and Balance Record

The basic empty mass (BEM) and center of gravity (CG) location are determined through actual weighing carried out by the helicopter manufacturer. This data is then entered on the first line of the MASS AND BALANCE RECORD, Form MBR-1 which then becomes a permanent part of the Flight Manual.

The MASS AND BALANCE RECORD must be updated (normally by transcribing the applicable information from the EQUIPMENT LIST) when necessary as follows:

- When additional equipment is installed on the helicopter necessitating a change in the basic empty mass (as per definition), add the new entry or entries to the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).
- Likewise, when equipment is removed from the helicopter, subtract the new entry or entries from the previous totals of basic empty mass and moment then compute the new basic empty mass, moment and CG location (arm).

At all times, the last mass and moment entries are considered the current basic empty mass and balance status of the helicopter.

6.2.2 Equipment List

An EQUIPMENT LIST, Form EL-1 is appended to this section and contains optional equipment of the particular helicopter when delivered. Each item on the list is provided with a number and description for identification, together with its mass, arm and moment.

Those items of equipment that were installed when the particular helicopter was initially weighed are so indicated by a check mark (✓) in the “Initial Weighing” column. Therefore the mass, arm and moment of these items are included in the basic empty mass (BEM) data found on the MASS AND BALANCE RECORD, Form MBR-1.